

# SHUBHAM RODE

Embedded System Engineer

+91 8104863990 | rodes8757@gmail.com | LinkedIn | Portfolio | Ahilyanagar, India

## PROFESSIONAL SUMMARY

E&TC graduate driven by precision and technical problem-solving, with hands-on experience in embedded systems, microcontroller integration, and hardware-software diagnostics equipped to troubleshoot, maintain, and optimize complex electro-mechanical equipment in a fast-paced manufacturing environment.

## EDUCATIONS

**K.K. Wagh Institute of Engineering Education and Research** | Nashik

Nov 2022 - Jun 2026

B.Tech in Electronics & Telecommunication

- Relevant Coursework:** Microcontrollers & Microprocessors, Real-Time Operating Systems (RTOS), VLSI, Embedded Systems, Internet of Things (IoT)

## SKILLS

|                                        |                                                                                                     |
|----------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Languages :</b>                     | C, Embedded C, Python, C++                                                                          |
| <b>Microcontrollers &amp; Boards :</b> | ATmega128, STM32, ESP32, Raspberry Pi, PIC18F4550, Arduino, Mega 2560                               |
| <b>Embedded Tools &amp; IDEs :</b>     | STM32CubeIDE, MPLAB X, Arduino IDE, Proteus, MATLAB, Keil µVision, Atmel Studio, Simulink, SavvyCan |
| <b>Sensors &amp; Actuators :</b>       | BMP180, MPX4250, RFID, Ultrasonic, LDR, DHT11, Servo Motors, Stepper Motors (NEMA 17)               |
| <b>Communication Protocols :</b>       | UART, I <sup>2</sup> C, SPI, CAN, RF, Bluetooth, Wi-Fi                                              |
| <b>Soft Skills :</b>                   | Problem Solving, Project-based Learning, Team Collaboration, Adaptability                           |

## WORK EXPERIENCES

**Renewable Energy & EV Operations Coordinator** | Internship

Jan 2026 - May 2026

Aviro Energy India Pvt Ltd

On-site • Pune

Led system-level layout and infrastructure design for EV charging stations and solar power grids. Conducted technical site analysis to identify optimal hardware placement, power routing, and load distribution for high-voltage setups. Coordinated the integration of EVSE (Electric Vehicle Supply Equipment) components and solar inverters, managing diagnostic data and project workflows through the I-SMART Portal and Bitrix24 CRM to enhance vehicle-to-grid power networks.

**Technologies / Skills Used :** Infrastructure Layout Design, EVSE Integration, Power Routing, High-Voltage Safety, Bitrix24 CRM

**Embedded System Engineer Intern** | Internship

Dec 2024 - Jan 2025

Samarth Solutions

Remote • NIT Bhopal

Led the comprehensive electromechanical design and assembly of a custom 3D printer, integrating an Arduino Mega 2560 and RAMPS 1.4 shield to achieve high dimensional print precision of  $\pm 0.1$  mm. Conducted rigorous hardware calibration of NEMA17 stepper motors and A4988 drivers to optimize multi-axis movement and mechanical load distribution. Coordinated hardware integration with Marlin firmware, tuning heat bed PID settings to reduce thermal fluctuations by 15% and ensure system stability for continuous fabrication cycles of up to 11 hours.

**Technologies / Skills Used :** Arduino Mega 2560, Marlin Firmware, RAMPS 1.4, NEMA17, MK8 Extruder, PLA Filament

## PROJECTS

**CAN-Sense: ECU Diagnostic and Traffic Monitor**

Apr 2026 - May 2026

System designer

Engineered a real-time CAN bus diagnostic monitor utilizing MCP2515 controllers to intercept ECU traffic at 500 kbps. Successfully decoded up to 1,000 frames per second with 99.9% logging accuracy.

**Technologies / Tools Used :** MCP2515 CAN Controller, STM32/ATmega128, SPI Protocol, CAN Bus (ISO 11898), Embedded C, ESP32, Arduino Uno, RaspberryPi 5

**NetBridge: Lightweight Layer 3 NAT Router**

Feb 2026 - Mar 2026

Firmware Developer

Engineered an ESP32-based Layer 3 NAT router to enhance Wi-Fi coverage by 35%. Achieved a consistent throughput of 5 Mbps for 5 concurrent devices and reduced packet loss by 25% through lwIP stack optimization.

**Technologies / Tools Used :** ESP32, lwIP Stack, TCP/IP, Network Address Translation (NAT), DHCP, Wi-Fi 802.11 b/g/n

**BharatVote - Smart Facial Recognition-Based Voting System**

Jul 2025 - Oct 2025

Team leader

Engineered a Raspberry Pi smart voting system using Python and OpenCV, achieving 98.5% authentication accuracy and verifying voters in under 1.2 seconds by reducing processing latency by 40%.

**Technologies / Tools Used :** Raspberry Pi 5, OpenCV, Face recognition, Thonny, Python 3.11, I2C protocol

**TireGuard - Integrated Nitrogen Tank System**

Sep 2024 - Oct 2024

Team Leader

Engineered an automated nitrogen gas inflation system utilizing an ATmega128 and MPX4250 sensors. Triggered solenoid valves in under 50 milliseconds to maintain tire pressure with  $\pm 0.5$  PSI precision.

**Technologies / Tools Used :** ATmega128, Embedded C, MPX4250 Pressure Sensor, 10-bit ADC, Solenoid Valve, Nitrogen Gas, Atmel Studio, NRF24L01